=== GNB (1780946120) ===


```yaml
# This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
# This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10 MHz bandwidth.
# To run the srsRAN Project gNB with this config, use the following command:
#   sudo ./gnb -c gnb_zmq.yaml

cu_cp:
  amf:
    addr: 10.53.1.2               # The address or hostname of the AMF.
    port: 38412
    bind_addr: 10.53.1.3            # A local IP that the gNB binds to for traffic from the AMF.
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "21407"
            tai_slice_support_list:
              - sst: 1
  inactivity_timer: 7200            # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 7200].

ru_sdr:
  device_driver: zmq                # The RF driver name.
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass arguments to the selected RF driver.
  srate: 11.52                      # RF sample rate might need to be adjusted according to selected bandwidth.
  tx_gain: 75                       # Transmit gain of the RF might need to adjusted to the given situation.
  rx_gain: 75                       # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:
  dl_arfcn: 368500                  # ARFCN of the downlink carrier (center frequency).
  band: 3                           # The NR band.
  channel_bandwidth_MHz: 10         # Bandwith in MHz. Number of PRBs will be automatically derived.
  common_scs: 15                    # Subcarrier spacing in kHz used for data.
  ssb_scs: 15                       # Subcarrier spacing in kHz used for SSB.
  plmn: "21407"                     # PLMN broadcasted by the gNB.
  tac: 7                            # Tracking area code (needs to match the core configuration).
  pdcch:
    common:
      ss0_index: 0                  # Set search space zero index to match srsUE capabilities
      coreset0_index: 6             # Set search CORESET Zero index to match srsUE capabilities
    dedicated:
      ss2_type: common              # Search Space type, has to be set to common
      dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
  prach:
    prach_config_index: 1           # Sets PRACH config to match what is expected by srsUE
  pdsch:
    mcs_table: qam64                # Sets PDSCH MCS to 64 QAM
  pusch:
    mcs_table: qam64                # Sets PUSCH MCS to 64 QAM

log:
  filename: /tmp/gnb.log            # Path of the log file.
  all_level: info                   # Logging level applied to all layers.
  hex_max_size: 0

pcap:
  mac_enable: false                 # Set to true to enable MAC-layer PCAPs.
  mac_filename: /tmp/gnb_mac.pcap   # Path where the MAC PCAP is stored.
  ngap_enable: false                # Set to true to enable NGAP PCAPs.
  ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
  e2ap_enable: true                 # Set to true to enable E2AP PCAPs.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap       # Path where the DU E2AP PCAP is stored.
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap  # Path where the CU-CP E2AP PCAP is stored.
```

```
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap  # Path where the CU-UP E2AP PCAP is stored.

#e2:
#  enable_du_e2: true                # Enable DU E2 agent (one for each DU instance)
#  e2sm_kpm_enabled: true            # Enable KPM service module
#  e2sm_rc_enabled: true             # Enable RC service module
 # addr: 127.0.0.1                   # RIC IP address
   #bind_addr: 127.0.0.100             # A local IP that the E2 agent binds to for traffic from the RIC. ONLY required if running the RIC on a separate machine.
#  port: 36421                       # RIC port

metrics:
  layers:
    enable_rlc: true                # Enable RLC metrics
    enable_sched: true              # Enable DU scheduler metrics
  periodicity:
    du_report_period: 1000          # Set DU statistics report period to 1s
```

=== UE (1780946120) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6


[rat.eutra]
dl_earfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 3
nof_carriers = 1

max_nof_prb = 106
nof_prb = 106


[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc  = 11223344556677889900aabbccddeeff00
k    = 00112233445566778899aabbccddeeff
imsi = 214070000000001
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1780946120) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
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# the LICENSE file in the top-level directory of this distribution
# and at http://www.gnu.org/licenses/.
#

services:
  5gc:
    container_name: open5gs_5gc
    build:
      context: open5gs
      target: open5gs
      args:
        OS_VERSION: "22.04"
        OPEN5GS_VERSION: "v2.7.0"
    environment:
      - MONGODB_IP=127.0.0.1
      - OPEN5GS_IP=10.53.1.2
      - UE_IP_BASE=10.45.0
      - UPF_ADVERTISE_IP=10.53.1.2
      - DEBUG=false
      - SUBSCRIBER_DB=214070000000001,00112233445566778899aabbccddeeff,opc,112233445566778899aabbccddeeff00,8000,9,10.45.1.2
      - NETWORK_NAME_FULL=srsRAN_ES
      - NETWORK_NAME_SHORT=srsRAN_ES
      - TZ=Europe/Madrid
    privileged: true
    ports:
      - "9898:9999/tcp"
    # Uncomment port to use the 5gc from outside the docker network
#      - "38412:38412/sctp"
 #     - "2152:2152/udp"
    command: 5gc -c open5gs-5gc.yml
    healthcheck:
      test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
      interval: 3s
      timeout: 1s
      retries: 60
    networks:
      ran:
        ipv4_address: ${OPEN5GS_IP:-10.53.1.2}

  gnb:
    container_name: srsran_gnb
    # Build info
    image: srsran/gnb
    build:
      context: ..
      dockerfile: docker/Dockerfile
```

```
      args:
        OS_VERSION: "24.04"
    # privileged mode is requred only for accessing usb devices
    privileged: true
    # Extra capabilities always required
    cap_add:
      - SYS_NICE
      - CAP_SYS_PTRACE
    volumes:
      # Access USB to use some SDRs
      - /dev/bus/usb/:/dev/bus/usb/
      # Sharing images between the host and the pod.
      # It's also possible to download the images inside the pod
      - /usr/share/uhd/images:/usr/share/uhd/images
      # Save logs and more into gnb-storage
      - gnb-storage:/tmp
    # It creates a file/folder into /config_name inside the container
    # Its content would be the value of the file used to create the config
    configs:
      - gnb_config.yml
      - gnb_compose_config.yml
    # Customize your desired network mode.
    # current netowrk configuration creastes a private netwoek with both containers attached
    # An alterantive would be `network: host"`. That would expose your host network into the container. It's the easiest to use if the 5gc is not in your PC
    networks:
      ran:
        ipv4_address: ${GNB_IP:-10.53.1.3}
      metrics:
        ipv4_address: 172.19.1.3
    # Start GNB container after 5gc is up and running
    depends_on:
      5gc:
        condition: service_healthy
    # Command to run into the final container
    command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
  gnb_config.yml:
    content: |
      # This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
      # This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10 MHz bandwidth.
      # To run the srsRAN Project gNB with this config, use the following command:
      #   sudo ./gnb -c gnb_zmq.yaml

      cu_cp:
        amf:
          addr: 10.53.1.2               # The address or hostname of the AMF.
          port: 38412
          bind_addr: 10.53.1.3            # A local IP that the gNB binds to for traffic from the AMF.
          supported_tracking_areas:
            - tac: 7
              plmn_list:
                - plmn: "21407"
                  tai_slice_support_list:
                    - sst: 1
          inactivity_timer: 7200             # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 7200].

      ru_sdr:
        device_driver: zmq                # The RF driver name.
        device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass arguments to the selected RF driver.
        srate: 11.52                         # RF sample rate might need to be adjusted according to selected bandwidth.
        tx_gain: 75                            # Transmit gain of the RF might need to adjusted to the given
```

```
situation.
         rx_gain: 75                           # Receive gain of the RF might need to adjusted to the given
situation.

      cell_cfg:
        dl_arfcn: 368500                  # ARFCN of the downlink carrier (center frequency).
        band: 3                           # The NR band.
        channel_bandwidth_MHz: 10         # Bandwith in MHz. Number of PRBs will be automatically derived.
        common_scs: 15                    # Subcarrier spacing in kHz used for data.
        ssb_scs: 15                       # Subcarrier spacing in kHz used for SSB.
        plmn: "21407"                     # PLMN broadcasted by the gNB.
        tac: 7                            # Tracking area code (needs to match the core configuration).
        pdcch:
          common:
            ss0_index: 0                  # Set search space zero index to match srsUE capabilities
            coreset0_index: 6             # Set search CORESET Zero index to match srsUE capabilities
          dedicated:
            ss2_type: common              # Search Space type, has to be set to common
            dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
        prach:
          prach_config_index: 1           # Sets PRACH config to match what is expected by srsUE
        pdsch:
          mcs_table: qam64                # Sets PDSCH MCS to 64 QAM
        pusch:
          mcs_table: qam64                # Sets PUSCH MCS to 64 QAM

      log:
        filename: /tmp/gnb.log            # Path of the log file.
        all_level: info                   # Logging level applied to all layers.
        hex_max_size: 0

      pcap:
        mac_enable: false                 # Set to true to enable MAC-layer PCAPs.
        mac_filename: /tmp/gnb_mac.pcap   # Path where the MAC PCAP is stored.
        ngap_enable: false                # Set to true to enable NGAP PCAPs.
        ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
        e2ap_enable: true                 # Set to true to enable E2AP PCAPs.
        e2ap_du_filename: /tmp/gnb_du_e2ap.pcap        # Path where the DU E2AP PCAP is stored.
        e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap  # Path where the CU-CP E2AP PCAP is stored.
        e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap  # Path where the CU-UP E2AP PCAP is stored.

      #e2:
      #  enable_du_e2: true                # Enable DU E2 agent (one for each DU instance)
      #  e2sm_kpm_enabled: true            # Enable KPM service module
      #  e2sm_rc_enabled: true             # Enable RC service module
       # addr: 127.0.0.1                   # RIC IP address
        #bind_addr: 127.0.0.100             # A local IP that the E2 agent binds to for traffic from the RIC.
ONLY required if running the RIC on a separate machine.
      #  port: 36421                       # RIC port

      metrics:
        layers:
          enable_rlc: true                # Enable RLC metrics
          enable_sched: true              # Enable DU scheduler metrics
        periodicity:
          du_report_period: 1000          # Set DU statistics report period to 1s
  gnb_compose_config.yml:
    content: |
      cu_cp:
        amf:
          addr: ${OPEN5GS_IP:-10.53.1.2}
          bind_addr: 0.0.0.0
      metrics:
        autostart_stdout_metrics: true
        enable_json: true
      remote_control:
        bind_addr: 0.0.0.0
        enabled: true
```

```yaml
volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24
```

=== NOTAS (1780946120) ===

1. PLMN configurado como "21407" (España) en gNB, UE (IMSI) y Core (SUBSCRIBER_DB) (REGLA 1, 7a).
2. ARFCN 368500 para Banda 3 es consistente (REGLA 2).
3. Se añadió `ssb_scs: 15` bajo `common_scs: 15` en gNB (REGLA 3).
4. Puertos ZMQ cruzados correctamente entre gNB y UE (REGLA 4).
5. `bind_addr` en `gnb_compose_config.yml` se estableció a `0.0.0.0` (REGLA 5).
6. Bloque `ru_sdr` copiado inmutablemente; srate 11.52 MHz es compatible con 10 MHz de ancho de banda (REGLA 6a, 6b, 6c).
7. IMSI, K y OPC son coherentes entre UE y SUBSCRIBER_DB, siguiendo el formato estricto (REGLA 7a, 7b).